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APL Photonics 10, 056110 (2025)

<https://doi.org/10.1063/5.0264068>



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Cite as: APL Photon. 10, 056110 (2025); doi: 10.1063/5.0264068

Submitted: 8 February 2025 • Accepted: 27 April 2025 •

Published Online: 12 May 2025



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ABSTRACT

We propose and demonstrate a broadband two-dimensional multi-beam optical phased array (MOPA) with four beams based on a Butler matrix on a silicon nitride (Si_3N_4) platform. A beamforming network utilizing cascaded Butler matrix splitting arrays has been proposed to scale up the MOPA for the first time, to the best of our knowledge, significantly reducing the number of waveguide crossings. A 32-channel MOPA with four beams has been designed and fabricated with a uniform antenna pitch of $3.5\ \mu\text{m}$. Experimental results indicate that the field of view is $25.6^\circ \times 8.1^\circ$, with a beam divergence of $0.655^\circ \times 0.085^\circ$. Simultaneous multi-beam 2D high-power operation and independently scanning along the wavelength steering axis have been demonstrated. Furthermore, parallel and high-speed free-space optical communication based on the fabricated MOPA has been performed. Data transmission of 40 Gbps non-return-to-zero on-off keying has been achieved, revealing the potential of low system complexity and power-efficient multi-access data transmission.

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I. INTRODUCTION

With the rapid development of photonic integrated circuits (PICs), chip-scale optical phased arrays (OPAs) are gaining increasing attention. By manipulating the phase and amplitude of optical antennas, OPAs enable precise and efficient beam steering. Compared to the electronic circuits, OPAs offer higher speed and lower power consumption, making them key technologies in applications such as light detection and ranging (LiDAR),^{1–3} free-space optical (FSO) communication,^{4–6} and image projection.⁷

Multi-beam optical phased arrays (MOPAs) enhance the functionality of single-beam OPAs by generating and controlling multiple laser beams simultaneously. This approach divides the field of view (FOV) into several parallel scanning regions, effectively multiplying the frame rate or enabling simultaneous communication of multiple terminals.

In recent years, many investigations have been reported on spatially duplicating the sub-arrays on a chip,^{8–10} cascading in free space¹¹ or using beamforming networks (BFNs).^{12–15} The first two methods trade off system complexity for multi-beam operation, which constrains the ultimate performance at a given level of integration. For the BFN method, in the field of radio frequency, multi-beam antennas have been at the forefront of modern wireless communications for decades.^{16–21} The Blass matrix, Nolen matrix, and Butler matrix (BM) are considered the most frequently used circuit-based BFNs. These BFNs are built from fundamental components such as couplers, splitters, and phase shifters (PSs). Due to their compatible fabrication process and cost-effectiveness, the BM is particularly prevalent for enabling phased array antenna systems.

OPAs based on BM designs have been demonstrated solely on the silicon-on-insulator (SOI) platform. However, for multi-access FSO systems, the silicon nitride (Si_3N_4) waveguide is more

suitable for high-power operation as it is unaffected by two-photon absorption and free-carrier absorption since its $\chi^{(3)}$ is ~ 20 times lower than that of silicon in the telecommunication wavelength range.^{22,23} Si_3N_4 also supports broadband operation and the realization of long grating antennas through single step etching. Moreover, large-scale antenna arrays can be used to achieve a sufficiently large optical aperture and to reduce the divergence angle of the beam radiated from the OPA chips, further increasing the steering resolution. However, directly increasing the scale of BFN based on BM will introduce considerable waveguide crossings. The number of crossings per channel increases exponentially at each stage, resulting in non-negligible on-chip loss and significantly limiting the level of integration.

In this paper, we demonstrate a broadband two-dimensional (2D) MOPA on the Si_3N_4 platform, utilizing a novel BFN formed by cascaded Butler matrix splitting arrays (BMSAs). With this proposed BFN, the number of crossings per channel at each stage is dramatically reduced from $2^{S+\log_2 M-1} - 1$ to a constant value of $M - 1$, where M denotes the number of input ports in the BM and S denotes the stage number. A 32-channel MOPA with four beams has been designed and fabricated, and multi-beam operation has been experimentally characterized. Independent operation of four beams along the wavelength scanning axis has been experimentally demonstrated. The FOV is measured to be $25.6^\circ \times 8.1^\circ$ with the divergence of $0.68^\circ \times 0.07^\circ$. We also demonstrate 40 Gbps non-return-to-zero (NRZ) on-off keying (OOK) data transmission on the MOPA-based FSO system. Clear eye diagrams of NRZ signals from four beams under the same phase shift voltage, as well as from a single beam at four different horizontal steering angles, have been demonstrated. It reveals the capability of high data rate transmission for the system.

II. STRUCTURE DESIGN AND CHARACTERIZATION

Figure 1 illustrates the schematic of the proposed MOPA, which consists of a 4×4 BM, a 4×32 BMSA formed by cascaded 4×8 BMSAs, a spiral phase shifter group array, and grating antennas.

By injecting light into port Nos. 1–4, beams 1–4 can be generated, respectively. Input light from the fiber array is coupled through the edge couplers into the input ports of BM. In the BM, the optical power is evenly distributed among the four output ports with a progressive phase difference (PPD). These four output ports form a group, and by passing through the cascade BMSAs, this group is duplicated into 2^N groups while maintaining the same PPD within, where N denotes the number of splitting stages of the BFN. The phase of the light in each group is then tuned by the thermo-optic PSs, enabling the light to be emitted into free space through the grating antenna array with the desired wavefront. To achieve multi-beam operation at different wavelengths simultaneously as well as broadband operation with a single wavelength calibration, the optical paths for all channels are designed to be equal.

In particular, the power consumption of the PS on the Si_3N_4 platform is relatively higher, and the bending radius is larger on the Si_3N_4 platform compared to the SOI platform. This is due to the thermo-optic coefficient of Si_3N_4 being approximately five times lower than that of silicon and the refractive index of Si_3N_4 ($n \approx 1.98$) being lower than that of Si ($n \approx 3.48$). Therefore, all the phase shifters in the MOPA are optimized with a spiral structure to reduce the power consumption,^{24–26} and the bends in PS are optimized using Bezier curves^{27,28} to achieve low loss and a small footprint. The widths of the three spiral waveguides are set to 0.9, 1.1, and 0.8 μm , respectively, to reduce crosstalk and minimize the footprint of the PS.

A. Butler matrix

Figure 2 shows the schematic of the BM. It consists of four multi-mode interference (MMI) couplers, two PSs, and two waveguide crossings. The 2×2 MMI couplers function as 90° hybrids, evenly splitting the optical power into two output ports while introducing a 90° phase difference to the cross-port. After the first stage of MMIs, PS_1 and PS_2 are adjusted to create a 45° phase difference relative to the crossing optical path. Finally, through the second stage of MMI couplers and a crossing, the optical power is divided into

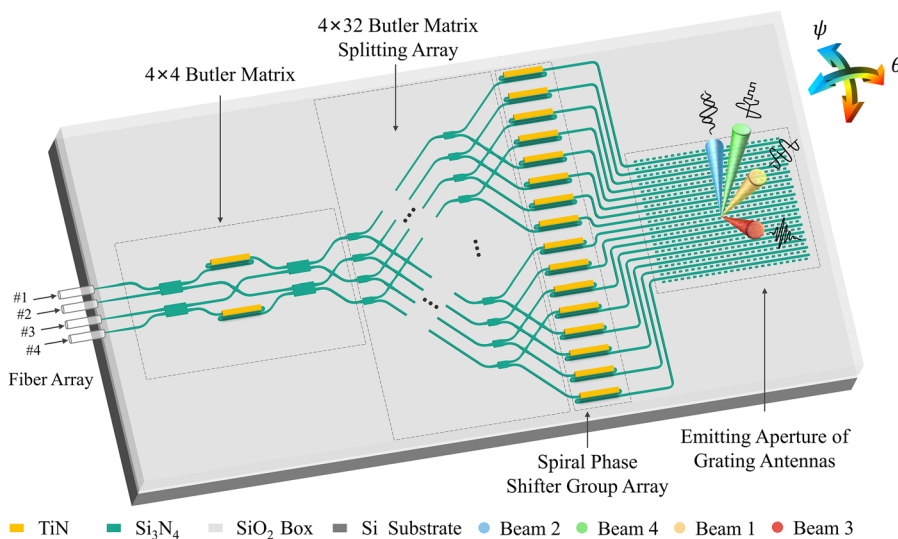


FIG. 1. Schematic of the proposed multi-beam optical phase array with four beams on a silicon-nitride platform.

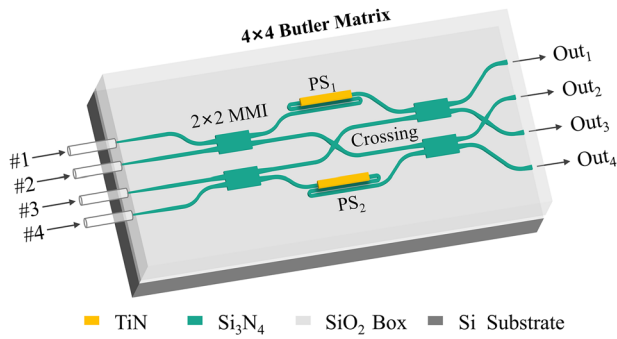


FIG. 2. Schematic of the Butler matrix.

four output ports, designated Out₁ to Out₄, each exhibiting a PPD of $\Delta\phi_B$ corresponding to the input port. The PPD response of different input ports is listed in Table I.

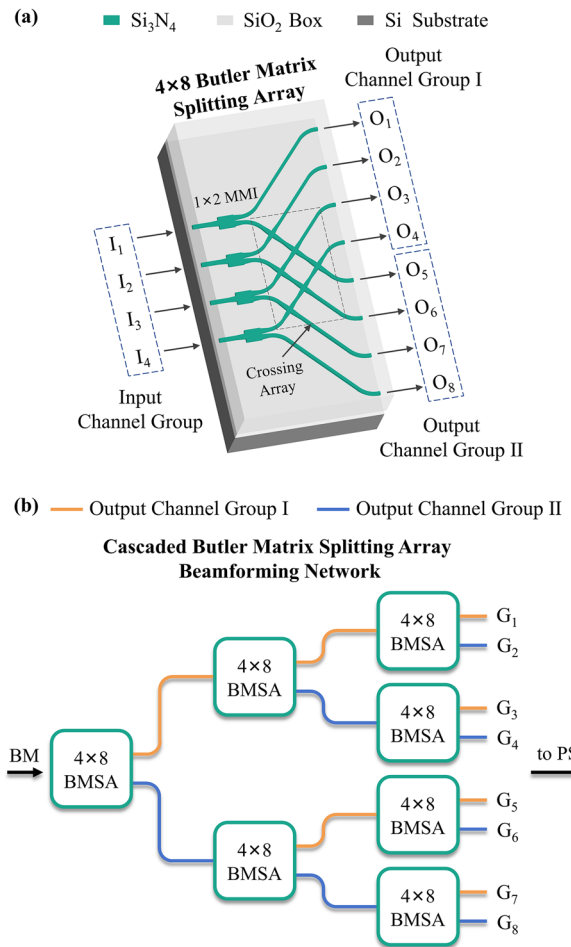
The 2×2 MMI couplers in the BM are designed based on the symmetrical interference for low-loss and broadband operation.^{29,30} The calculated splitting ratio for the MMI coupler is within 0.49–0.51, and excess loss is <0.5 dB over the 100 nm wavelength band (from 1500 to 1600 nm). The deviation in the 90° phase difference introduced by the MMI coupler is less than 1.26° over the whole band, which is negligible. The waveguide crossing is designed utilizing shaped taper optimization,³¹ achieving an insertion loss of less than 0.05 dB and a crosstalk of less than -50 dB. The edge coupler is designed with a 300 nm tip, and the coupling efficiency is higher than 95%. In addition, the insertion loss of PS is calculated to be 0.95 dB within the bandwidth of 100 nm. Compared with the BM on the SOI platform, these devices can perform with a broader wavelength band while maintaining low loss and phase error.

B. Cascaded Butler matrix splitting array beamforming network

Figure 3(a) shows the schematic of the proposed 4×8 BMSA. The BMSA consists of four MMI couplers and a waveguide crossing array with only three crossings for each channel since the optical path is compensated. The output channels are divided into two groups (group I with channels O₁–O₄ and group II with channels O₅–O₈), while I₁–I₄ are the input channel group. Figure 3(b) illustrates the schematic of the proposed cascaded BMSA BFN at the channel group scale. In this context, the proposed 4×8 BMSA functions as a 1×2 splitter, featuring two distinct output channel groups, with output channel group I indicated by yellow lines

TABLE I. Phase distribution among each output port for different input ports (rad).

Port	No. 1	No. 2	No. 3	No. 4
Out ₁	$\pi/4$	$3\pi/4$	$\pi/2$	π
Out ₂	$\pi/2$	0	$5\pi/4$	$3\pi/4$
Out ₃	$3\pi/4$	$5\pi/4$	0	$\pi/2$
Out ₄	π	$\pi/2$	$3\pi/4$	$\pi/4$
$\Delta\phi_B$	$\pi/4$	$-3\pi/4$	$3\pi/4$	$-\pi/4$

FIG. 3. (a) Schematic of the proposed 4×8 BMSA. (b) Schematic of the proposed 4×32 BMSA (cascaded 4×8 BMSA) beamforming network.

and output channel group II indicated by blue lines. By cascading the BMSA across three stages, the output channel group of the BM is effectively replicated into eight groups, labeled as G₁–G₈. Within each channel group, the PPD maintains $\Delta\phi_B$. However, the phase differences among channels from different groups do not align with the intended PPD. To address this issue, the PSs before the emitting aperture are organized into groups corresponding to the channel groups. A progressive phase shift bias of $\phi_G = \pi$ is introduced between adjacent groups, which is equivalent to applying a phase shift bias to all PS groups following output channel group II of the final stage.

Table II provides an example of the phase distribution among each output channel from a 4×8 BMSA in the final BFN stage when input No. 1 is activated. Initially, before tuning by the PS group array, both output channel groups exhibit the same phase distribution as that of the BM. Subsequently, with an additional π bias applied to the channels in output channel group II, the PS groups introduce a progressive phase shift of ϕ , enabling simultaneous steering of four beams along the phase array scanning axis

TABLE II. Phase distribution among each output channel from a final stage 4×8 BMSA, before and after the PS groups, along with the phase shift of the PS for the corresponding output channels (rad).

Group type	Output channel group I				Output channel group II			
Channel	O ₁	O ₂	O ₃	O ₄	O ₅	O ₆	O ₇	O ₈
Before PS groups	$\pi/4$	$\pi/2$	$3\pi/4$	π	$\pi/4$	$\pi/2$	$3\pi/4$	π
Phase shift	φ	2φ	3φ	4φ	$5\varphi + \pi$	$6\varphi + \pi$	$7\varphi + \pi$	$8\varphi + \pi$
After PS groups	$\varphi + \pi/4$	$2\varphi + \pi/2$	$3\varphi + 3\pi/4$	$4\varphi + \pi$	$5\varphi + 5\pi/4$	$6\varphi + 3\pi/2$	$7\varphi + 7\pi/4$	$8\varphi + 2\pi$

ψ . Ultimately, the desired phase distribution is achieved, as illustrated in the last row of Table II. The number of channels, denoted as 2^P , is a crucial parameter for enhancing the performance of OPAs. The total number of waveguide crossings through each channel is defined as N . Previously reported MOPAs introduce an exponentially increasing number of waveguide crossings, resulting in $N = 2^P - P - 2$. In contrast, our proposed BFN introduces a constant value of waveguide crossings per stage; the number N is only $3(P - 2)$. For example, when $2^P = 1024$, N is only 24 using our proposed BFN, whereas N is 1012 with the previously reported approach. This significant reduction in N indicates that the cascaded BMSA BFN can dramatically reduce the complexity and, consequently, the on-chip loss of the entire MOPA chip.

The PPD $\Delta\varphi_B$ can be summed up as

$$\Delta\varphi_B = (2n - 3) \times \frac{\pi}{4}, n \in \{0, 1, 2, 3\}. \quad (1)$$

Assuming that k is the number of BMSA input ports, the number of crossings per channel per stage is $(k - 1)$. To increase the number of channels while compensating for phase differences among channels from different groups to meet the intended PPD, φ_G must satisfy the following expression:

$$\varphi_G = k \times \Delta\varphi_B \bmod 2\pi. \quad (2)$$

Find the minimum value of k such that expression (2) holds for any $n \in \{0, 1, 2, 3\}$; therefore,

$$2nk \times \frac{\pi}{4} = 0 \bmod 2\pi. \quad (3)$$

The minimum value of k is 4. In this case, $\varphi_G = \pi$. Therefore, the proposed cascaded BMSA BFN has the theoretically minimum number of crossings.

C. Grating antennas

The Si₃N₄ platform has been recognized as suitable for achieving a millimeter-scale effective antenna length.³² In this work, we propose an antenna with a millimeter-scale effective length, consisting of fully etched subwavelength gratings. Figure 4(a) shows the schematic of the proposed grating antennas. The waveguide width is tapered from 900 nm to 1.1 μ m before the grating region to slightly reduce the grating strength. On both sides of the waveguide, with a 250 nm gap, 2500 subwavelength gratings are defined at a period of 1200 nm spanning over a length of 3 mm. In each period, the subwavelength strip is 350 nm in width by 600 nm in length. Due to the $\pi/2$ phase interval among the PPDs generated by different inputs, as referenced in Table I, four beams are emitted through the shared aperture at equal angular intervals of 6.4° within the aliasing-free FOV of $\pm 12.8^\circ$, as shown in Fig. 4(b). Considering the variation in emitting strength and the constructive interference among the grating periods, the beam steering angle and full width at

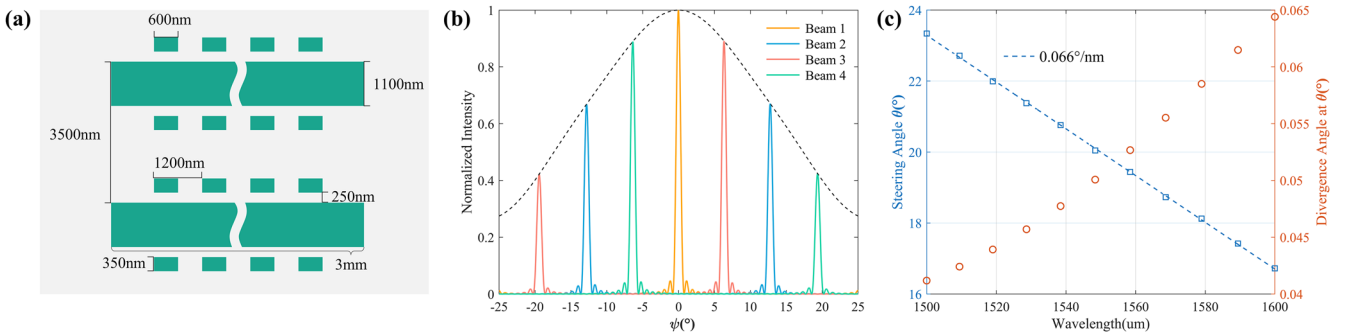


FIG. 4. (a) Schematic of the proposed grating antennas. (b) Calculated far-field intensity distribution of the multi-beam pattern at an operating wavelength of $1.55 \mu\text{m}$, with beam 1 calibrated to 0° of phase array scanning axis ψ . The black dashed line represents the beam-steering envelope. (c) Scanning efficiency retrieved through the linear regression of the steering angle of wavelength scanning axis θ and the corresponding divergence angle.

half maximum (FWHM) divergence profiles of far-field spots at different wavelengths are calculated from 1500 to 1600 nm, as shown in Fig. 4(c).

III. FABRICATION AND CHARACTERIZATION

The MOPA is then fabricated on the 400 nm Si_3N_4 platform with a 4 μm buried oxide layer and a 2 μm oxide cladding layer. Figure 5(a) shows the fabricated and packaged MOPA, with a footprint of $7.8 \times 5.3 \text{ mm}^2$. The Si_3N_4 layer is deposited by the low-pressure chemical vapor deposition and flattened through chemical mechanical polishing. The patterns are defined by the 248 nm deep UV (DUV) lithography followed by a dry etching process. The distance between the adjacent phase shifters is set to 65 μm . TiN micro-heaters are connected to pads through aluminum wires and then wire-bonded to a printed circuit board (PCB).

The measured splitting ratio for the MMI coupler is within 0.51–0.55, and the loss is <0.9 dB over 80 nm wavelength band (from 1530 to 1610 nm). The measured insertion loss and crosstalk for the waveguide crossing are less than 0.085 dB and –38 dB, respectively. The maximum insertion loss of the spiral PS and edge coupler is

1.2 and 1.3 dB, respectively. In addition, the measured diffraction strength for the antennas is 2.7–3.3 dB/mm.

A multi-channel voltage source is connected to the PCB through the I/O ports, controlling the phase shift of each channel independently. A feedback loop is established, consisting of a voltage source, a MOPA chip, an infrared CCD camera, and a computer. In the initial calibration step, a modified rotating element vector algorithm³³ is employed to compensate for the initial random phase error and ϕ_G with only input No. 1 activated. The figure of merit (FOM) is defined as the mainlobe power. By optimizing the driving voltages of the PSs in each channel using this algorithm, the FOM converges to its maximum value, indicating that the random phase errors have been effectively compensated. The total power consumption is about 1.92 W. In the subsequent step, the phase relationship between the BM and the PS group array is utilized to calibrate the PSs within the BM with a second input activated. By iterating these steps, the performance of the MOPA can be further optimized. In our demonstration, only a single calibration round is required due to the inherent robustness of the MOPA phase distribution. After calibration, the CCD camera captures the far-field distribution over a fixed FOV.

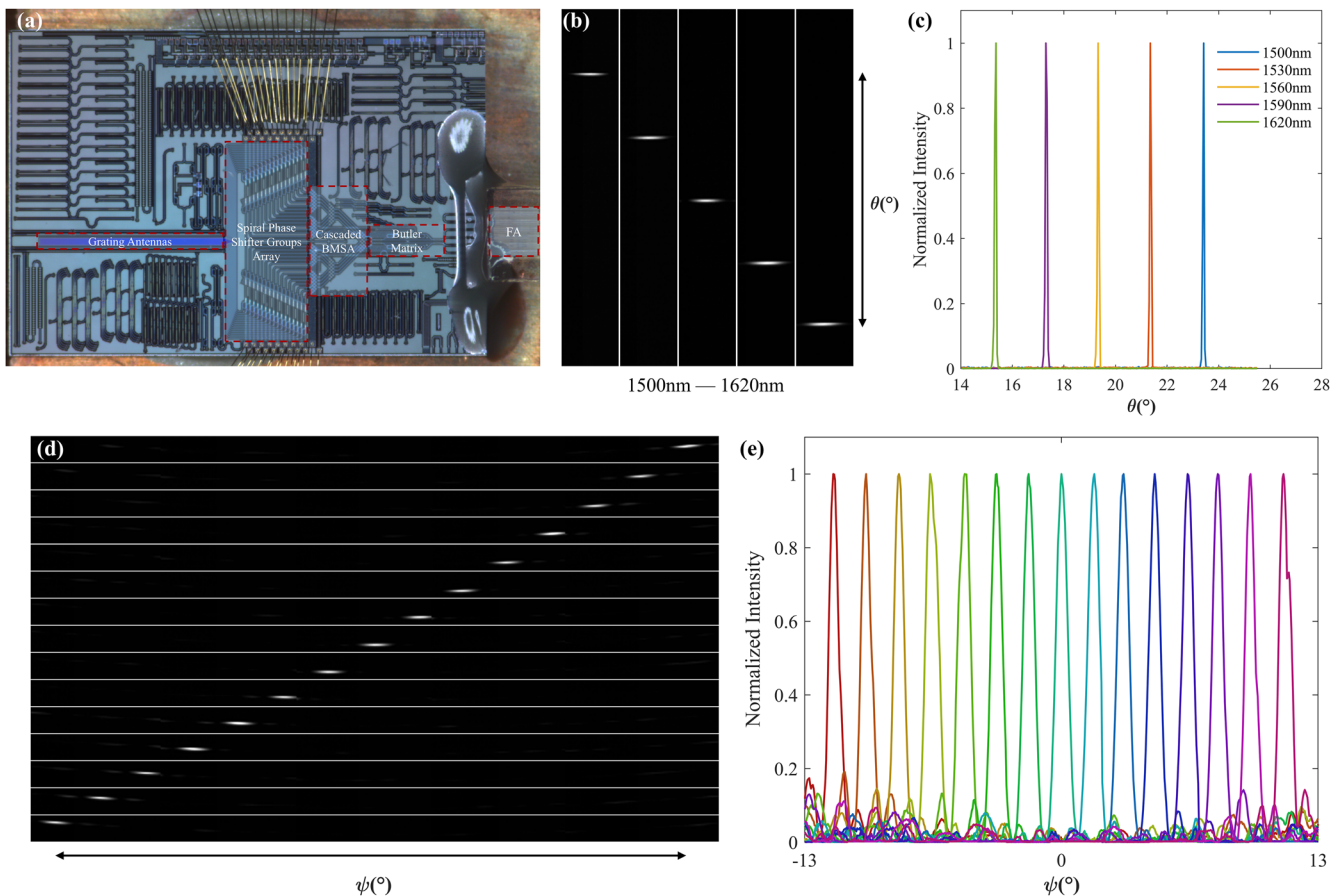


FIG. 5. (a) Microscope image of the fabricated and packaged multi-beam optical phase array chip. (b) Far-field beam images along the wavelength scanning axis θ . (c) Measured far-field distribution at wavelengths from 1500 to 1620 nm with 30 nm interval. (d) Far-field beam images along the phase array scanning axis ψ . (e) Corresponding far-field distribution at steering angles from -11.2° to 11.2° with 1.6° interval.

To characterize the 2D beam steering ability of the shared aperture, we tune the phase shifters and wavelength while activating input No. 1. The CCD camera is rotated to capture the far-field distribution. By tuning the wavelength, beam 1 can be steered along axis θ . As shown in Figs. 5(b) and 5(c), we realize a beam steering range of 8.1° by changing the wavelength from 1500 to 1620 nm, which closely matches the calculated scanning efficiency. At the wavelength of 1550 nm, we measure the far-field distribution when the beam is steering along axis ψ , as shown in Fig. 5(d). The measured far-field distribution at different steering angles is shown in Fig. 5(e). The sidelobe suppression ratio (SLSR) is observed to be higher than 7.2 dB within the FOV, which is generally consistent with the simulation results. The SLSR can be improved by increasing the rounds of calibration or manipulating the amplitude distribution of the antenna array.^{34,35} The average beam divergence is measured to be $0.655^\circ \times 0.085^\circ$ along axis ψ and θ . The measured beam divergence is consistent with the calculated values.

To characterize multi-beam 2D operation, a Fourier imaging system is employed to capture the multi-beam images. Figure 6(a) illustrates the schematic of the multi-beam imaging system. Each of the four tunable lasers is connected to a polarization controller, ensuring TE polarization operation. The multi-channel voltage source is used to simultaneously steer the four beams. The near-field multi-beam pattern is captured and transferred to the far-field pattern on the focal plane through a microscope objective (MO). This far-field pattern is then projected onto the CCD camera using two lenses (lens 1: $f = 5$ cm, lens 2: $f = 15$ cm). The FOV of this Fourier imaging system is $\sim 80^\circ$. As previously mentioned, the four beams emitted by the MOPA scan in parallel along the phase array scanning axis ψ . Since the aliasing free FOV of MOPA is $\pm 12.8^\circ$,

grating lobes can be observed in the captured images. We steer beam 1 to -6.4° , -3.2° , 0° , 3.2° , and 6.4° , with all the four input wavelengths set at 1550 nm, as shown in Fig. 6(b). The four beams within the aliasing-free FOV are denoted by red marks corresponding to their input port numbers. Along the wavelength scanning axis, the four beams operate independently. To demonstrate this characteristic, the input wavelengths of beam 3 and beam 4 are set to 1520 and 1580 nm, respectively. As shown in Fig. 6(c), compared to the condition where all four beams operate at 1550 nm, the far-field patterns of beam 1 and beam 2 remain fixed in their initial positions within the image. Meanwhile, the far-field patterns of beam 3 and beam 4 exhibit a vertical shift, resulting in a sawtooth-like distribution pattern in the far-field images of the four beams, which indicates the function of arbitrary beamforming generation. This characteristic is attributed to the optimized equal optical path design and wavelength transparency of the proposed MOPA on the Si_3N_4 platform.

Since the beams are only parallel along the axis ψ , while other characteristics remain independent, the system exhibits sufficient flexibility. In addition, the proposed MOPA is well-suited for high-power operation. As a proof of concept, a high-speed FSO communication system is experimentally demonstrated using the proposed MOPA as the transmitter and a collimator as the receiver.

As shown in Fig. 7(a), the experiment setup consists of tunable laser sources, an arbitrary waveform generator, a commercial lithium niobate Mach-Zehnder modulator (LN-MZM), a driver, a direct current source, two polarization controllers (PCs), two erbium-doped fiber amplifiers (EDFAs), an optical filter, and a high-speed oscilloscope. Four tunable lasers are used for the MOPA input signals. Input light first passes through PC No. 1 and is then modulated by the LN-MZM, generating a 40 Gbps NRZ OOK data stream.

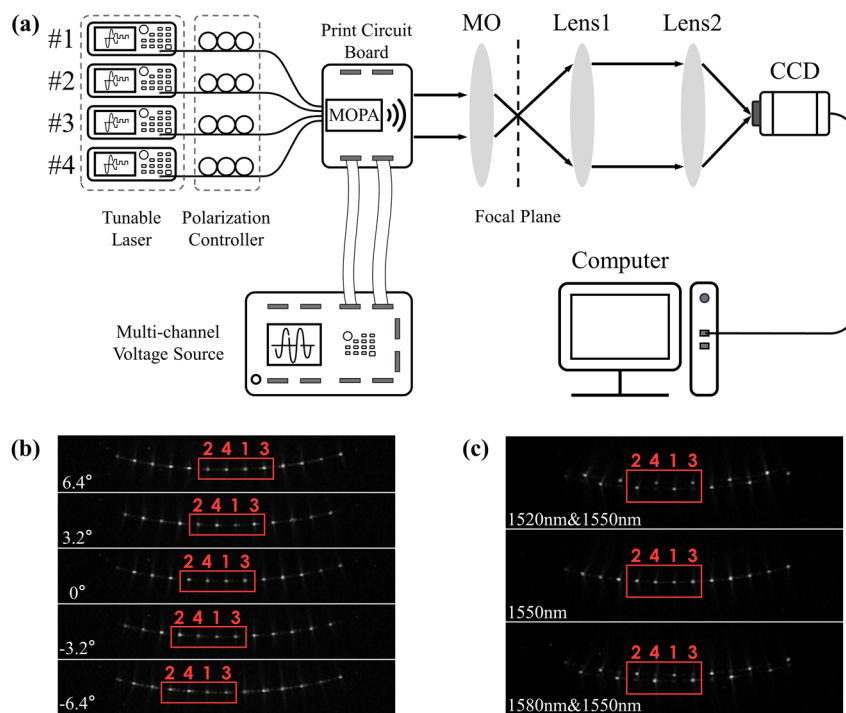


FIG. 6. (a) Schematic of the multi-beam imaging system. (b) Measured multi-beam images when beam 1 is steered to -6.4° , -3.2° , 0° , 3.2° , and 6.4° , with all the input wavelengths set at 1550 nm. (c) Measured multi-beam patterns with the input wavelengths of Nos. 3 and 4 at 1520, 1550, and 1580 nm, while Nos. 1 and 2 at 1550 nm.

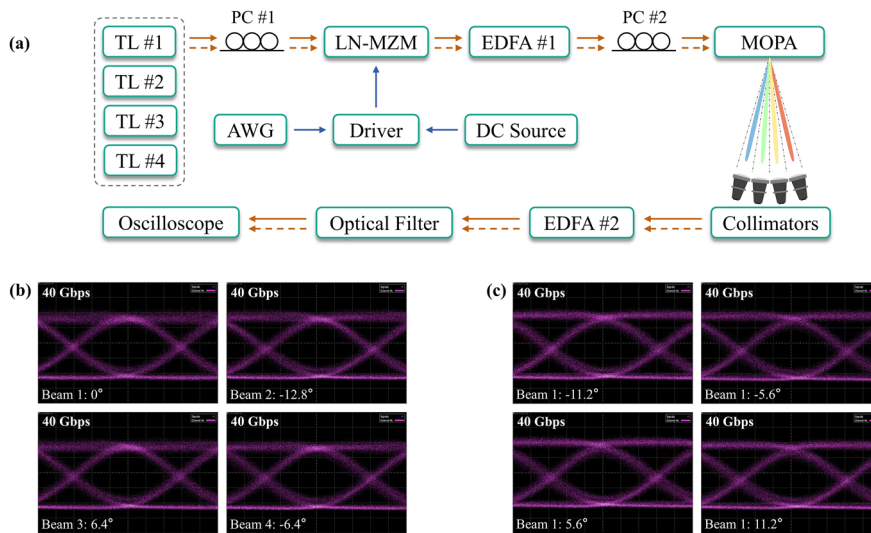


FIG. 7. (a) Schematic of the experiment setup for a free-space optical communication system based on a multi-beam optical phase array. (b) Eye diagrams of NRZ signals from beam 1 to beam 4 under the same phase shift voltage. (c) Eye diagrams of NRZ signals from beam 1 at four different steering angles in the phase array scanning axis ψ .

The modulated signal is subsequently amplified to 0.25 W by EDFA No. 1. The amplified signal then passes through PC No. 2 and is coupled into the MOPA through the input ports, after which it is emitted into the free space and received by the different collimators. Finally, the signal is amplified by EDFA No. 2 and collected by the oscilloscope for analysis.

Data transmission through different beams, as well as through a single beam at different steering angles, has been demonstrated. As previously mentioned, when beam 1 is directed to 0° , beams 2, 3, and 4 are at -12.8° , 6.4° , and -6.4° , respectively. As shown in Fig. 7(b), the same signal with NRZ OOK data is transmitted by these four beams. The measured extinction ratios (ERs) are 8.88, 10.08, 9.49, and 10.04 dB, while the measured signal-to-noise ratios (SNRs) are 8.89, 8.85, 9.04, and 9.05 dB. The data quality is highly consistent among the four beams, as they only share the beamforming optical path. Figure 7(c) shows the measured eye diagrams of beam 1 when it is steered to -11.2° , -5.6° , 5.6° , and 11.2° . The eye diagrams are open with the measured ERs of 9.91, 10.17, 9.99, and 10.17 dB and the measured SNRs of 8.42, 8.17, 7.73, and 7.47 dB. It can be seen that the measured eye diagrams of the four beams are clear and distinct, indicating the capability of parallel high-speed data transmission with multi-beam operation.

IV. CONCLUSION

In summary, we propose and demonstrate a two-dimensional MOPA with four beams based on a BM on a Si_3N_4 platform. Compared to the prior arts, the novel BFN consists of cascaded BMSAs, which reduces the waveguide crossing number through each channel from an exponentially growing number of $2^P - P - 2$ to $3(P - 2)$. The MOPA forms multiple beams, which are capable of 2D high-power operation and independently scanning along the wavelength steering axis. Table III shows the performance comparison with previous studies. In addition, we have demonstrated high-speed FSO communication based on the proposed MOPA chip. 40 Gbps NRZ OOK data transmission with distinct eye diagrams has been observed for all the four beams, indicating the BM based MOPA an ideal candidate for achieving both low system complexity and power-efficient multi-access data transmission. The FOV of MOPA is measured to be $25.6^\circ \times 8.1^\circ$ with the divergence of $0.68^\circ \times 0.07^\circ$. The main limitation on the FOV arises from the relatively large antenna pitch of Si_3N_4 -based antennas, which is caused by strong coupling between adjacent waveguides, particularly when the antenna length exceeds 1 mm. By introducing silicon PSs and antennas on the Si_3N_4 -on-Si platform, both power consumption and FOV

TABLE III. Performance comparison with previous studies.

	Ref. 8	Ref. 10	Ref. 11	Ref. 15	This work
Number of beams	2	4	4	4	4
Multi-beamforming method	Sub-arrays	Sub-arrays	Amplitude-phase cascade	Conventional BM-BFN	Novel BM-BFN
Number of waveguide crossings ^a	...	...	...	$2^P - P - 2$	$3(P - 2)$
Integration platform	...	Si_3N_4	Liquid crystal	SOI	Si_3N_4
SLSR ^b	5.2 dB	...	<1.9 dB	<5.5 dB	7.2 dB
Divergence angle	6°	$4.3^\circ \times 0.7^\circ$	...	$3.2^\circ \times 3.91^\circ$	$0.655^\circ \times 0.085^\circ$
FOV	50°	$17.6^\circ \times 3^\circ$	...	$60^\circ \times 8^\circ$	$25.6^\circ \times 8.1^\circ$

^aWithin each channel; 2^P represents the number of OPA channels.

^bThe worst in FOV.

can be further improved. Owing to their higher optical confinement, silicon waveguides enable smaller antenna pitch. In addition, using the upper Si_3N_4 layer as a grating perturbation allows antenna lengths to exceed 1 cm.

ACKNOWLEDGMENTS

The authors acknowledge the National Major Research and Development Program (No. 2021YFB2801703), National Natural Science Foundation of China (Grant Nos. 62135011, 62105286, and 62335001), "Pioneer" and "Leading Goose" R&D Program of Zhejiang (Grant No. 2022C01103), Leading Innovative and Entrepreneur Team Introduction Program of Zhejiang (Grant No. 2021R01001), and the Fundamental Research Funds for the Central Universities.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

W.J. and J.C. contributed equally to this work.

Wenke Jiao: Formal analysis (lead); Investigation (lead); Methodology (lead); Writing – original draft (lead); Writing – review & editing (equal). **Jingye Chen:** Investigation (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal). **Daixin Lian:** Formal analysis (equal); Investigation (equal); Methodology (equal); Writing – original draft (equal). **Hongxuan Liu:** Investigation (equal); Methodology (equal); Writing – original draft (equal). **Mingyu Zhu:** Investigation (equal); Methodology (equal); Writing – original draft (equal). **Shi Zhao:** Investigation (equal); Project administration (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal). **Daixin Dai:** Funding acquisition (equal); Investigation (equal); Supervision (equal); Writing – review & editing (equal). **Yaocheng Shi:** Funding acquisition (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

- C. V. Poulton, M. J. Byrd, P. Russo, E. Timurdogan, M. Khandaker, D. Vermeulen, and M. R. Watts, "Long-range LiDAR and free-space data communication with high-performance optical phased arrays," *IEEE J. Sel. Top. Quantum Electron.* **25**, 7700108 (2019).
- J. K. Doylend and S. Gupta, "An overview of silicon photonics for LiDAR," *Proc. SPIE* **11285**, 112850J (2020).
- B. Chen, Y. Li, Q. Xie, Q. Na, M. Tao, Z. Wang, Z. Zhi, H. Hu, X. Li, H. Qu, Y. He, X. Hu, G. Lo, and J. Song, "SiN-on-SOI optical phased array LiDAR for ultra-wide field of view and 4D sensing," *Laser Photonics Rev.* **18**, 2301360 (2024).
- H.-W. Rhee, J.-B. You, H. Yoon, K. Han, M. Kim, B. G. Lee, S.-C. Kim, and H.-H. Park, "32 Gbps data transmission with 2D beam-steering using a silicon optical phased array," *IEEE Photonics Technol. Lett.* **32**, 803–806 (2020).
- Y. Li, B. Chen, Q. Na, M. Tao, X. Liu, Z. Zhi, T. Peng, X. Li, X. Luo, G.-Q. Lo, F. Gao, X. Li, Q. Xie, and J. Song, "High-data-rate and wide-steering-range optical wireless communication via nonuniform-space optical phased array," *J. Lightwave Technol.* **41**, 4933–4940 (2023).
- K. Wang, A. Nirmalathas, C. Lim, E. Wong, K. Alameh, H. Li, and E. Skafidas, "High-speed indoor optical wireless communication system employing a silicon integrated photonic circuit," *Opt. Lett.* **43**, 3132–3135 (2018).
- M. Raval, A. Yaacobi, and M. R. Watts, "Integrated visible light phased array system for autostereoscopic image projection," *Opt. Lett.* **43**, 3678–3681 (2018).
- Y. Wu, S. Shao, and D. Che, "Fast and low grating lobe multi-beam steering with a subarray level unequally spaced optical phased array," *J. Opt. Soc. Am. B* **38**, 3417–3424 (2021).
- Y. Liu, Z. Hao, L. Wang, J. Wang, J. Yu, B. Xiong, C. Sun, H. Li, Y. Han, and Y. Luo, "On-chip multi-beam emitting optical phased array for wide-angle LiDAR," in *Conference on Lasers and Electro-Optics (2020)* (Optica Publishing Group, 2020) Paper JTu2G.25.
- N. A. Tyler, D. Fowler, S. Malhouitre, S. Garcia, P. Grosse, W. Rabaud, and B. Szlag, "SiN integrated optical phased arrays for two-dimensional beam steering at a single near-infrared wavelength," *Opt. Express* **27**, 5851–5858 (2019).
- L. Wu, X. Wang, X. He, Z. Huang, X. Huang, and C. Xiong, "Arbitrary multiple beam forming by two cascaded liquid crystal optical phased arrays," *Opt. Express* **26**, 17066–17077 (2018).
- C. Tsokos, C. Kouloumentas, E. Mylonas, P. Groumas, V. Katopodis, L. Gounaridis, R. B. Timens, R. M. Oldenbeuving, C. G. H. Roeloffzen, and H. Avramopoulos, "Analysis of a multibeam optical beamforming network based on blass matrix architecture," *J. Lightwave Technol.* **36**, 3354–3372 (2018).
- P. Lu, W. Xu, C. Zhu, C. Liu, L. Lu, L. Zhou, and J. Chen, "Integrated multi-beam optical phased array based on a 4×4 Butler matrix," *Opt. Lett.* **46**, 1566 (2021).
- Z. Zhou, W. Xu, X. Li, L. Lu, J. Chen, and L. Zhou, "Butler matrix enabled multi-beam optical phased array for two-dimensional beam-steering," in *2022 27th OptoElectronics and Communications Conference (OECC) and 2022 International Conference on Photonics in Switching and Computing (PSC)* (IEEE, 2022), pp. 1–3.
- Z. Zhou, W. Xu, C. Liu, R. Xu, C. Zhu, X. Li, L. Lu, J. Chen, and L. Zhou, "Butler matrix enabled multi-beam optical phased array for two-dimensional beam-steering and ranging," *Photonics Res.* **12**, 912 (2024).
- J. Butler, "Beam-forming matrix simplifies design of electronically scanned antennas," *Electron. Des.* **9**, 170–173 (1961).
- J. Blass, "Multidirectional antenna—A new approach to stacked beams," in *1958 IRE International Convention Record* (IEEE, 1960), Vol. 8, pp. 48–50.
- H. Ren, H. Zhang, Y. Jin, Y. Gu, and B. Arigong, "A novel 2-D 3×3 Nolen matrix for 2-D beamforming applications," *IEEE Trans. Microwave Theory Tech.* **67**, 4622–4631 (2019).
- X.-Z. Wang, F.-C. Chen, and Q.-X. Chu, "A compact broadband 4×4 Butler matrix with 360° continuous progressive phase shift," *IEEE Trans. Microwave Theory Tech.* **71**, 3906–3914 (2023).
- A. Tajik, A. Shafiei Alavijeh, and M. Fakharzadeh, "Asymmetrical 4×4 butler matrix and its application for single layer 8×8 Butler matrix," *IEEE Trans. Antennas Propag.* **67**, 5372–5379 (2019).
- S. Mosca, F. Bilotti, A. Toscano, and L. Vegni, "A novel design method for Blass matrix beam-forming networks," *IEEE Trans. Antennas Propag.* **50**, 225–232 (2002).
- M. Dinu, F. Quochi, and H. Garcia, "Third-order nonlinearities in silicon at telecom wavelengths," *Appl. Phys. Lett.* **82**, 2954 (2003).
- C. V. Poulton, M. J. Byrd, P. Russo, B. Moss, O. Shatrovov, M. Khandaker, and M. R. Watts, "Coherent LiDAR with an 8,192-element optical phased array and driving laser," *IEEE J. Sel. Top. Quantum Electron.* **28**, 508 (2022).

- ²⁴A. Densmore, S. Janz, R. Ma, J. H. Schmid, D.-X. Xu, A. Del  ge, J. Lapointe, M. Vachon, and P. Cheben, "Compact and low power thermo-optic switch using folded silicon waveguides," *Opt. Express* **17**, 10457–10465 (2009).
- ²⁵S. Zhao, J. Chen, D. Dai, and Y. Shi, "Low-loss and power-efficient phase shifter based on an optimized multimode spiral silicon waveguide," *Opt. Lett.* **48**, 4653 (2023).
- ²⁶Z. Yong, H. Chen, X. Luo, A. Govdeli, H. Chua, S. S. Azadeh, A. Stalmashonak, G.-Q. Lo, J. K. S. Poon, and W. D. Sacher, "Power-efficient silicon nitride thermo-optic phase shifters for visible light," *Opt. Express* **30**, 7225–7237 (2022).
- ²⁷C. Koos, C. G. Poulton, L. Zimmermann, L. Jacome, J. Leuthold, and W. Freude, "Ideal bend contour trajectories for single-mode operation of low-loss overmoded waveguides," *IEEE Photonics Technol. Lett.* **19**, 819–821 (2007).
- ²⁸W. Li, J. Chen, D. Liang, D. Dai, and Y. Shi, "Silicon optical phased array with calibration-free phase shifters," *Opt. Express* **30**, 44029 (2022).
- ²⁹L. B. Soldano and E. C. M. Pennings, "Optical multi-mode interference devices based on self-imaging: Principles and applications," *J. Lightwave Technol.* **13**, 615–627 (1995).
- ³⁰P. A. Besse, M. Bachmann, H. Melchior, L. B. Soldano, and M. K. Smit, "Optical bandwidth and fabrication tolerances of multimode interference couplers," *J. Lightwave Technol.* **12**, 1004–1009 (1994).
- ³¹A. M. Jones, C. T. DeRose, A. L. Lentine, D. C. Trotter, A. L. Starbuck, and R. A. Norwood, "Ultra-low crosstalk, CMOS compatible waveguide crossings for densely integrated photonic interconnection networks," *Opt. Express* **21**, 12002–12013 (2013).
- ³²Y. Shuai, Z. Zhou, and H. Su, "Toward practical optical phased arrays through grating antenna engineering," *Photonics* **10**, 520 (2023).
- ³³D. Lian, S. Zhao, W. Li, J. Chen, D. Dai, and Y. Shi, "Grating-lobe-free optical phased array with 2-D circular sparse array aperture and high-efficiency phase calibration," *Nanophotonics* **13**, 29–37 (2024).
- ³⁴Y. Liu and H. Hu, "Silicon optical phased array with a 180-degree field of view for 2D optical beam steering," *Optica* **9**, 903–907 (2022).
- ³⁵S. Zhao, D. Lian, W. Li, J. Chen, D. Dai, and Y. Shi, "Low sidelobe silicon optical phased array with Chebyshev amplitude distribution," *Nanophotonics* **13**, 263–269 (2024).